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(54) INHALER

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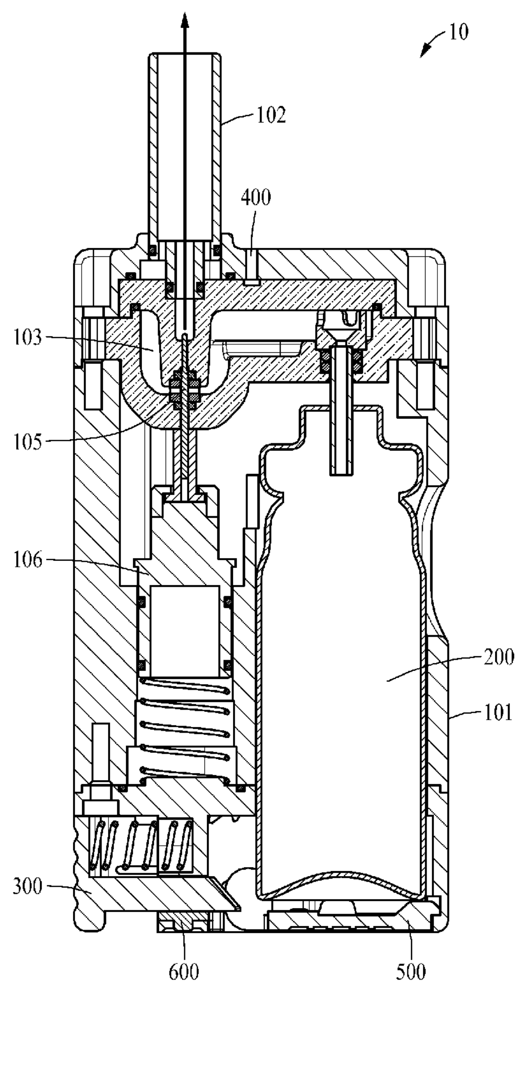
(57) **ABSTRACT**

An inhaler, according to one embodiment, comprises: a housing having one surface, the other surface opposite the one surface, and a plurality of side surfaces connecting the one surface and the other surface; a mouthpiece disposed on the one surface of the housing; a reservoir disposed within the housing and storing an inhalable composition; a nozzle extending from the mouthpiece to the reservoir; a needle valve movably disposed inside the nozzle and operating in a first state to close the nozzle or a second state to open the nozzle; and a sealing member fixedly installed on the nozzle, wherein a first gap may be formed between the needle valve and the nozzle in the first state, and a second gap may be formed between the needle valve and the sealing member in the second state.



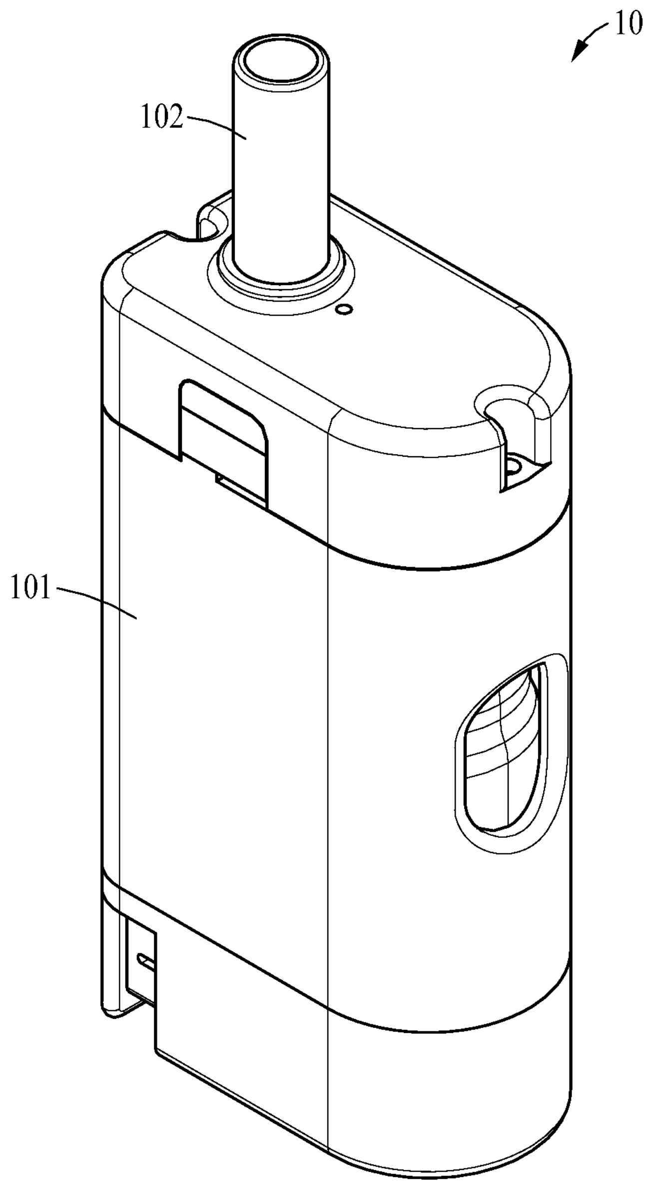


FIG. 1

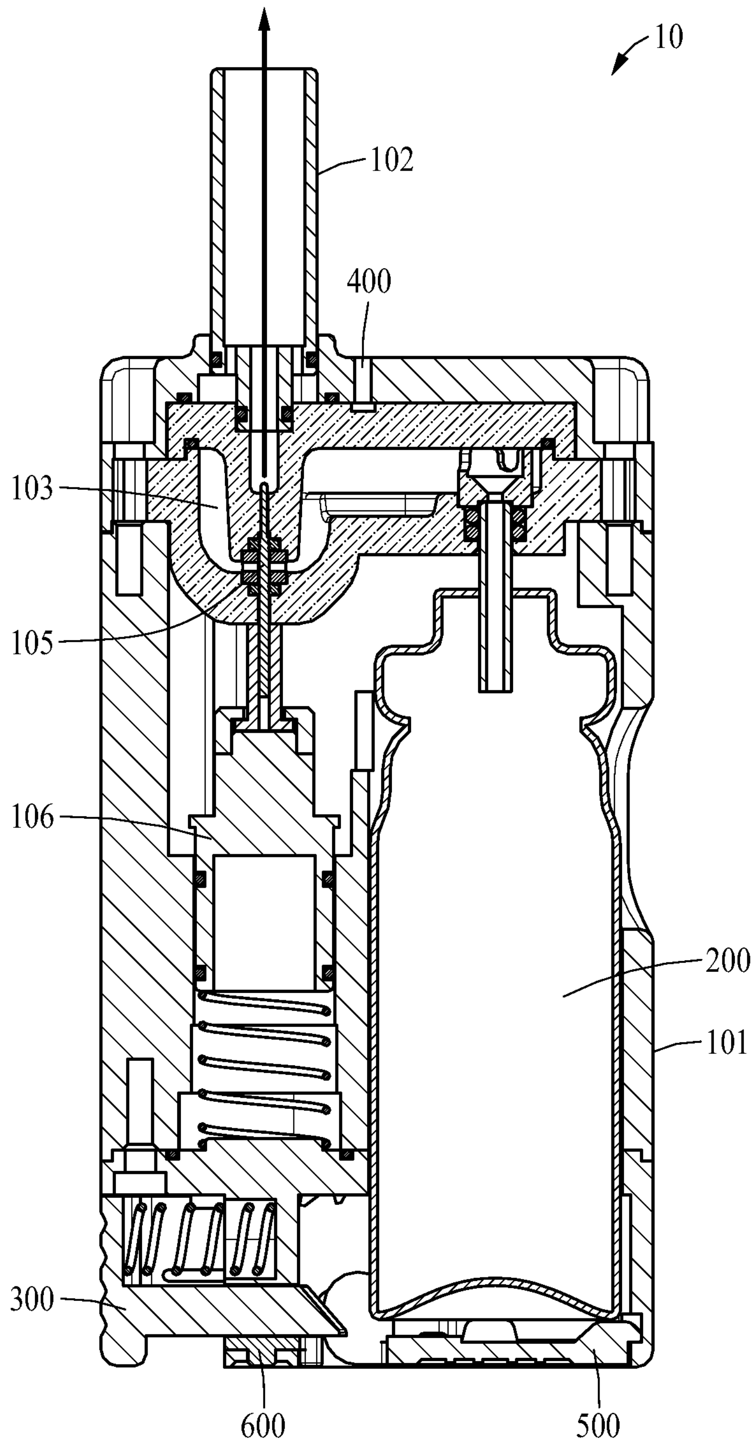


FIG. 2

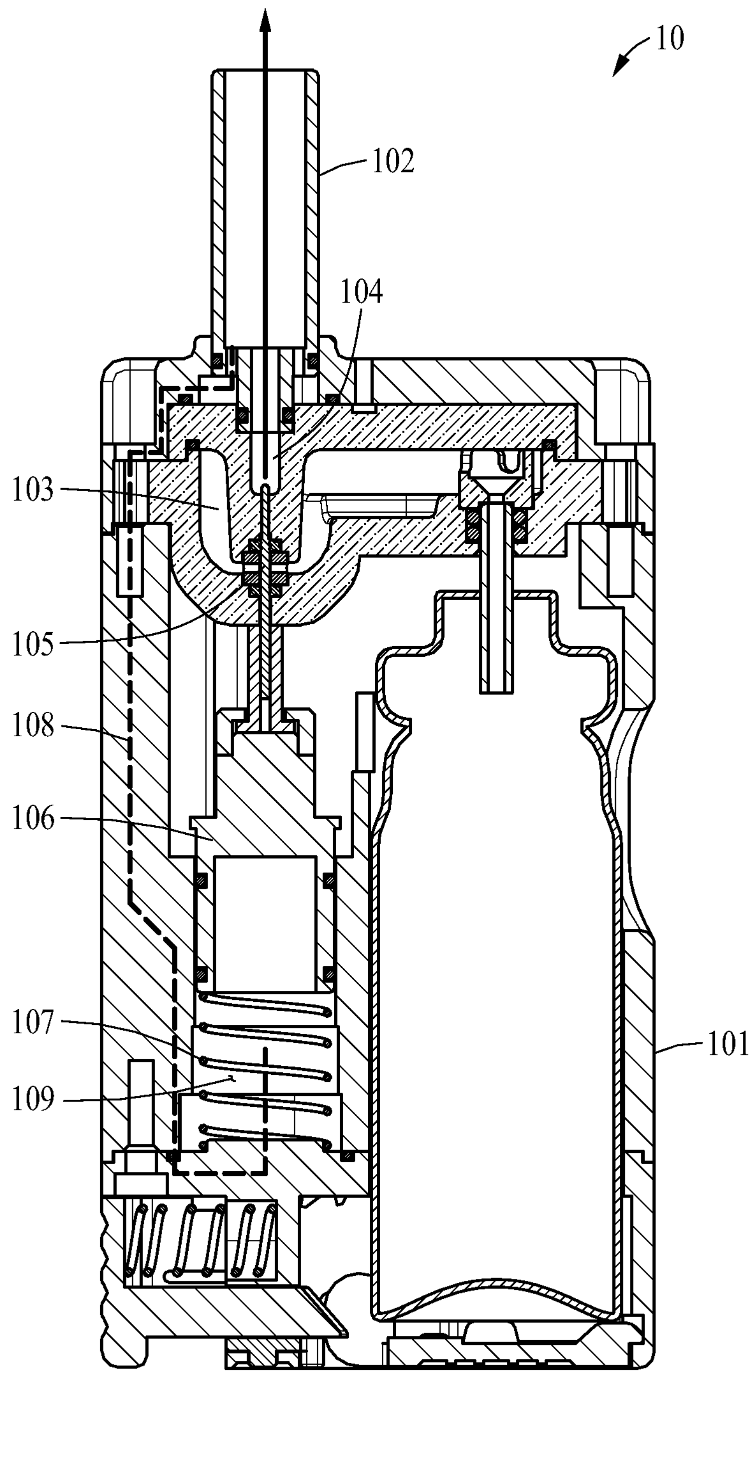


FIG. 3

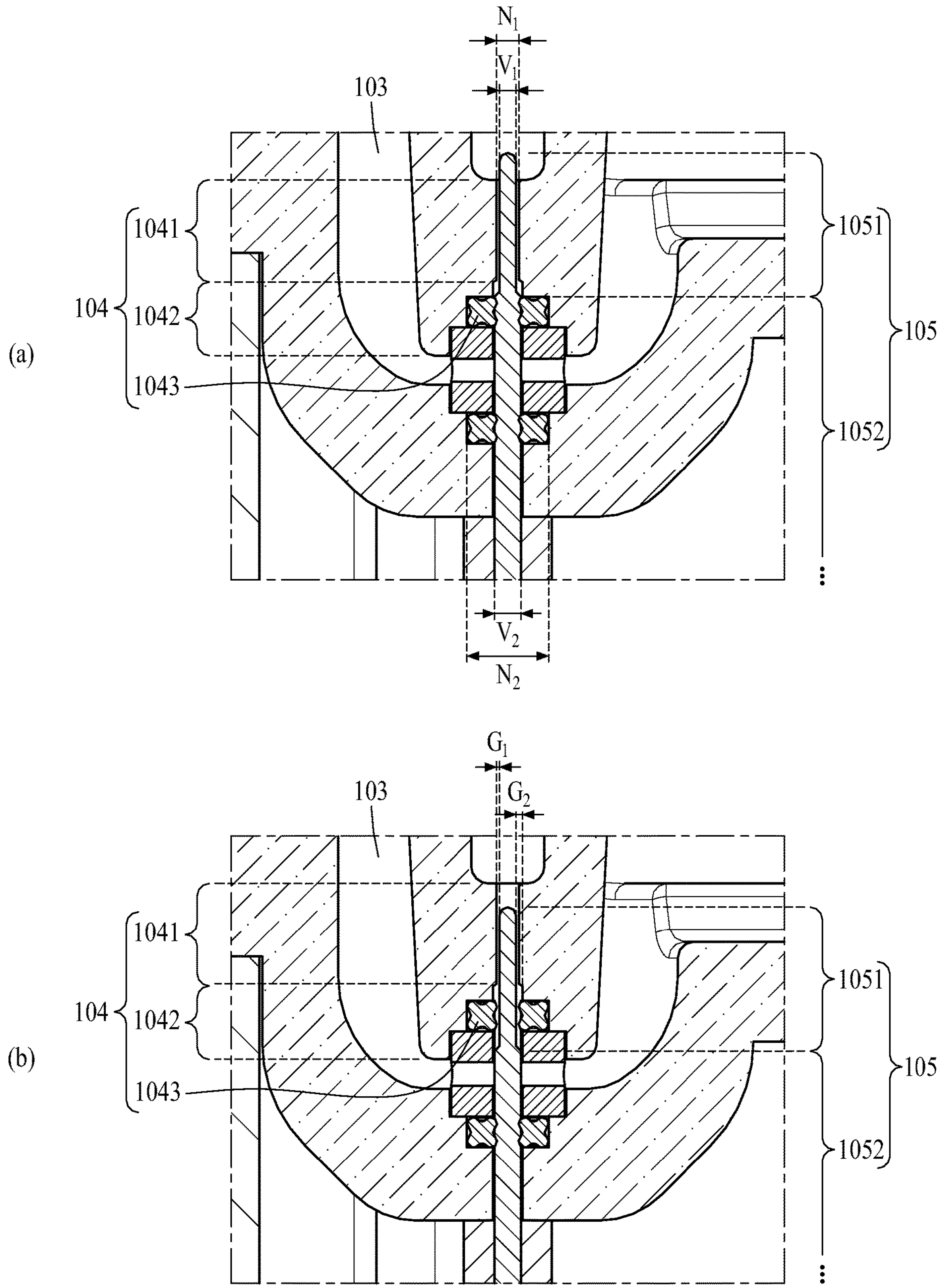

(a)
N1
V1
103
1041
1042
1043
104
1051
1052
105
V2
N2
(b)
G1
G2
103
1041
1042
1043
104
1051
1052
105

FIG. 4

INHALER

TECHNICAL FIELD

[0001] Disclosed is an inhaler.

BACKGROUND ART

[0002] In general, an inhaler is a device used to inhale a composition such as medication through the oral cavity or nasal cavity as a liquid or gas in the process of inhalation. Such an inhaler may include a container accommodating an inhalable composition, and the composition may be sprayed from the container through a thin tube to the oral cavity or nasal cavity through an intake to be inhaled by a user.

[0003] The above description has been possessed or acquired by the inventor(s) in the course of conceiving the present disclosure and is not necessarily an art publicly known before the present application is filed.

[0004] Prior Document: Korean Patent Gazette No. 10-1759972

DISCLOSURE OF THE INVENTION

Technical Goals

[0005] An object according to an embodiment is to provide an inhaler having a technology in which a flow path for ejecting an aerosol is divided into two spaces having different cross-sectional areas according to an operation of a valve and an appropriate section of the two divided spaces is specified and aerosol formation conditions are derived to enable uniform ejection of a sprayed aerosol.

[0006] The technical tasks obtainable from the present disclosure are non-limited by the above-mentioned technical tasks. And, other unmentioned technical tasks can be clearly understood from the following description by those having ordinary skill in the technical field to which the present disclosure pertains.

Technical Solutions

[0007] An inhaler according to an embodiment for achieving the above objects includes: a housing having one surface, the other surface opposite to the one surface, and a plurality of side surfaces connecting the one surface and the other surface; a mouthpiece which is disposed on the one surface of the housing; a reservoir which is disposed inside the housing and stores an inhalable composition; a nozzle which extends from the mouthpiece to the reservoir; a needle valve which is movably disposed inside the nozzle and operates in a first state for closing the nozzle and a second state for opening the nozzle; and a sealing member which is fixedly provided on the nozzle. In the first state, a first gap is formed between the needle valve and the nozzle, and in the second state, a second gap is formed between the needle valve and the sealing member.

[0008] According to an aspect, the first state may be defined as a state in which the needle valve comes into contact with the sealing member, and the second state may be defined as a state in which the needle valve does not come into contact with the sealing member.

[0009] According to an aspect, the first gap may be smaller than the second gap.

[0010] According to an aspect, the nozzle may include a first nozzle portion adjacent to the mouthpiece; and a second nozzle portion adjacent to the reservoir. The first nozzle portion may be formed to have a diameter smaller than a diameter of the second nozzle portion.

[0011] According to an aspect, the needle valve may include: a first valve portion which is formed on a front end and vertically reciprocates in a first direction from the other surface to the one surface of the housing or a second direction from the one surface to the other surface of the housing through the first nozzle portion and the second nozzle portion; and a second valve portion which is formed on a lower portion of the first valve portion. The first valve portion may be formed to have a diameter smaller than a diameter of the second valve portion.

[0012] According to an aspect, the first valve portion may be formed to have the diameter smaller than the diameter of the first nozzle portion.

[0013] According to an aspect, the first valve portion may not come into contact with the first nozzle portion in the first state or the second state, and the second valve portion may come into contact with the sealing member in the first state and may not come into contact with the sealing member in the second state.

[0014] According to an aspect, the first gap may be formed between the first valve portion and the first nozzle portion, and the second gap may be formed between the first valve portion and the sealing member.

[0015] According to an aspect, the second gap may allow movement of the inhalable composition discharged from the reservoir, and the first gap may allow the movement of the inhalable composition to the mouthpiece by controlling a particle size of the inhalable composition passing through the second gap.

[0016] According to an aspect, the first gap may be formed to 0.015 millimeters (mm) to 0.03 mm.

[0017] According to an aspect, the inhaler may further include: a piston which has one surface coupled to the needle valve, and vertically reciprocates in a first direction from the other surface to the one surface of the housing or a second direction from the one surface to the other surface of the housing; and a spring which is provided below the piston in a preloaded state. When a suction force is not applied through the mouthpiece, the spring may push the piston up in the first direction to maintain the first state, and when a suction force is applied through the mouthpiece, the piston may overcome preload of the spring and move in the second direction to switch the state into the second state.

[0018] According to an aspect, the inhaler may further include a passage which is formed inside the housing and extends from one side of the mouthpiece to a bottom portion of the piston. The passage may transfer the suction force applied to the mouthpiece to the piston.

[0019] According to an aspect, the inhaler may further include a negative pressure forming portion which forms a space between the piston and the passage. The negative pressure forming portion may induce the piston to move in the second direction by generating a negative pressure by the suction force.

Effects

[0020] According to the inhaler according to an embodiment, there are effects that a flow path for ejecting an aerosol is divided into two spaces having different cross-sectional areas according to an operation of a valve and an appropriate section of the two divided spaces is specified and aerosol formation conditions are derived to enable uniform ejection of a sprayed aerosol.

[0021] The effects of the inhaler are not limited to the above-mentioned effects, and other unmentioned effects can be clearly understood from the following description by one of ordinary skill in the art to which the present disclosure pertains.

BRIEF DESCRIPTION OF DRAWINGS

[0022] FIG. **1** is a perspective view of an inhaler according to an embodiment.

[0023] FIG. **2** is a cross-sectional view of an inhaler according to an embodiment.

[0024] FIG. **3** is a cross-sectional view of an inhaler according to an embodiment.

[0025] FIG. **4** illustrates an inhaler in a first state and a second state.

[0026] The accompanying drawings illustrate desired embodiments of the present disclosure and are provided together with the detailed description for better understanding of the technical idea of the present disclosure. Therefore, the present disclosure should not be construed as being limited to the embodiments set forth in the drawings.

BEST MODE FOR CARRYING OUT THE INVENTION

[0027] Hereinafter, embodiments will be described in detail with reference to the illustrative drawings. Regarding the reference numerals assigned to the components in the drawings, it should be noted that the same components will be designated by the same reference numerals, wherever possible, even though they are shown in different drawings. Further, in the following description of the present embodiments, a detailed description of publicly known configurations or functions incorporated herein will be omitted when it is determined that the detailed description obscures the subject matters of the present embodiments.

[0028] In addition, the terms first, second, A, B, (a), and (b) may be used to describe constituent elements of the embodiments. These terms are used only for the purpose of discriminating one constituent element from another constituent element, and the nature, the sequences, or the orders of the constituent elements are not limited by the terms. When one constituent element is described as being "connected", "coupled", or "attached" to another constituent element, it should be understood that one constituent element can be connected or attached directly to another constituent element, and an intervening constituent element can also be "connected", "coupled", or "attached" to the constituent elements.

[0029] The same name may be used to describe an element included in the embodiments described above and an element having a common function. Unless otherwise mentioned, the descriptions on the embodiments may be applicable to the following embodiments and thus, duplicated descriptions will be omitted for conciseness.

[0030] FIG. **1** is a perspective view of an inhaler **10** according to an embodiment.

[0031] FIG. **2** is a cross-sectional view of the inhaler **10** according to an embodiment.

[0032] FIG. **3** is a cross-sectional view of the inhaler **10** according to an embodiment.

[0033] FIG. **4** illustrates the inhaler **10** in a first state and a second state.

[0034] Referring to FIG. **1**, the inhaler **10** according to an embodiment may include a housing **101** and a mouthpiece **102**.

[0035] The housing **101** may include a first surface formed on one surface, a second surface opposite to the first surface, and a plurality of side surfaces connecting the first surface and the second surface. The first surface of the housing **101** may be, for example, a surface located on the top of the housing **101**, and the second surface may be, for example, a bottom surface of the housing **101**. Hereinafter, a direction from the second surface to the first surface is defined as a first direction, and a direction from the first surface to the second surface is defined as a second direction.

[0036] The mouthpiece **102** may be disposed on the first surface of the housing **101**. A user may inhale an inhalable composition accommodated in the inhaler **10** through the mouthpiece **102**. At this time, the user may inhale the composition, for example, in the form of an aerosol or in the form of a powder. Hereinafter, the inhaler **10** according to an embodiment will be described using the inhaler **10** that sprays an inhalable composition in the form of an aerosol as an example.

[0037] Referring to FIG. **2**, in the inhaler **10** according to an embodiment, a canister **200** accommodating an inhalable composition may be mounted on the inside of the housing **101**, and a certain amount of the composition of the canister **200** may be filled in a reservoir **103**. The user may visually identify a remaining amount of the composition stored in the reservoir **103** through a viewing window (not shown) provided on the housing **101**. A filling lever **300** that may be pressed by the user may be provided to fill **20** the inside of the reservoir **103** with the composition accommodated in the canister **200**. The filling lever **300** may be positioned on the second surface of the housing **101**, and when the user presses the filling lever **300**, the filling lever **300** may push up a bottom surface of the canister **200** in the first direction so that an injection hole of the canister **200** communicates with the reservoir **103**, and the composition may move from the canister **200** to the reservoir **103** through the injection hole. In addition, the inhaler **10** according to an embodiment may include the counter (not shown) that counts the number of times of filling in connection with a vertical movement of the canister **200** during the filling of the reservoir **103**, and a display window (not shown) that displays a remaining amount of the composition in the canister **200**. The composition stored in the reservoir **103** may be sprayed in the form of an aerosol as the user applies a suction force to the mouthpiece **102**. At this time, an inhalation interlocking valve **105** that opens and closes the reservoir **103** by the suction force may operate, and the opening and closing operation of the inhalation interlocking valve **105** may be controlled by a piston **106**. Meanwhile, a relief valve (not shown) may be provided on the reservoir **103**, and a relief vent hole **400** may be provided on the first surface of the housing **101**. The relief valve is interlocked with the filling lever **300** through a relief bar (not shown) or a relief bar movement protrusion (not shown), and accordingly, the relief valve may be opened first to discharge a residual gas in the reservoir **103**, before the filling lever **300** pushes the canister **200** up in the first direction to fill the reservoir **103**. The inhaler **10** according to an embodiment may further include a cover **500** and a locking device **600** for preventing the canister **200** from being separated.

[0038] In the inhaler **10** described above, a needle valve method developed to solve problems that occur when an aerosol is sprayed into the user's nasal cavity or oral cavity is applied to the inside of the housing **101**, and this needle valve method will be described in detail below with reference to FIGS. **3** and **4**.

[0039] In general, when a pinch valve method is applied to an inhaler, a problem of wetting the inside of the user's oral cavity may occur due to too large particles of the aerosol sprayed through a nozzle having a circular cross-sectional area. In addition, since a tip of an injection nozzle is positioned inside the oral cavity, a problem in that the aerosol hits the oral cavity hard during the spraying may occur. In order to solve such problems, the needle valve method is applied in the present disclosure.

[0040] Referring to FIG. **3**, the inhaler **10** according to an embodiment may further include the reservoir **103**, a nozzle **104**, and a needle valve **105**.

[0041] The reservoir **103** may be disposed inside the housing **101**. The reservoir **103** may store an inhalable composition.

[0042] The nozzle **104** may allow the mouthpiece **102** to communicate with the reservoir **103**. The nozzle **104** may be provided as a tube extending from the mouthpiece **102** to the reservoir **103**.

[0043] The needle valve **105** may be movably disposed inside the nozzle **104**. For example, the needle valve **105** may move up and down inside the nozzle **104**. The needle valve **105** may open or close the nozzle **104** depending on its position within the nozzle **104**. That is, the needle valve **105** may open the nozzle **104** so that the mouthpiece **102** and the reservoir **103** communicate with each other, or may close the nozzle **104** so that the mouthpiece **102** and the reservoir **103** are isolated from each other.

[0044] In the needle valve method using the needle valve **105**, as the needle-shaped valve **105** is positioned in the middle of the nozzle **104**, a cross-sectional area for spraying an aerosol may maintain a donut shape. In addition, finer particles may be formed by spraying the aerosol from a narrower section compared to the same injection area, and a phenomenon of hitting or wetting the inside of the oral cavity may be improved, because the nozzle **104** for spraying an aerosol is positioned at a lower tip of the mouthpiece **102** which is far from the oral cavity.

[0045] The nozzle **104** may be switched to one of a first state or a second state by the needle valve **105**. The first state is a state in which the nozzle **104** is closed, and the second state is a state in which the nozzle **104** is open. In the first state, the mouthpiece **102** and the reservoir **103** may be isolated from each other. In the second state, the mouthpiece **102** and the reservoir **103** may communicate with each other. That is, in the second state, the inhalable composition stored in the reservoir **103** may be discharged to the mouthpiece **102** through the nozzle **104** as indicated by an arrow.

[0046] In addition, the needle valve **105** may operate in association with suction.

[0047] Specifically, when a suction force is not applied to the mouthpiece **102**, the needle valve **105** may maintain the nozzle **104** in the first state. When a suction force is applied through the mouthpiece **102**, the needle valve **105** may move in the second direction to switch the nozzle **104** into the second state.

[0048] Referring back to FIG. **3**, the inhaler **10** according to an embodiment may further include the piston **106**, a spring **107**, a passage **108**, and a negative pressure forming portion **109**.

[0049] The piston **106** may be disposed inside the housing **101**, and one end of the needle valve **105** may be coupled to one surface of the piston **106**. The piston **106** may reciprocate vertically in a cylinder.

[0050] The spring **107** may be provided below the piston **106**. At this time, the spring **107** may be provided in a preloaded state.

[0051] The passage **108** may be formed inside the housing **101**. The passage **108** may allow the mouthpiece **102** to communicate with the piston **106**. Specifically, one end of the passage **108** may be connected to one side of the mouthpiece **102**, and the other end of the passage **108** may be connected to the bottom of the piston **106**. At this time, one end of the passage **108** may be connected to the mouthpiece **102** at a position spaced apart from a position where the nozzle **104** is connected to the mouthpiece **102**. The passage **108** may transfer a suction force applied to the mouthpiece **102** to the piston **106**.

[0052] The negative pressure forming portion **109** may be formed between the piston **106** and one end of the passage **108** connected to the piston **106**. The negative pressure forming portion **109** may form a space between the piston **106** and the passage **108**. When a suction force is applied to the mouthpiece **102**, the suction force transferred through the passage **108** may reach the negative pressure forming portion **109**, and the negative pressure forming portion **109** may generate a negative pressure. Accordingly, the negative pressure forming portion **109** may induce the piston **106** to move in the second direction.

[0053] As a result, when a suction force is not applied to the mouthpiece **102**, the spring **107** may push the piston **106** up in the first direction so that the needle valve **105** and the nozzle **104** are maintained in the first state. On the other hand, when a suction force is applied to the mouthpiece **102**, the suction force may be transferred to the negative pressure forming portion **109** through the passage **108**, and the piston **106** may overcome the preload of the spring **107** by the negative pressure generated by the negative pressure forming portion **109**, and move in the second direction. Accordingly, the needle valve **105** may move in the second direction and the nozzle **104** may be switched into the second state.

[0054] Hereinafter, referring to FIG. **4**, the nozzle **104** and the needle valve **105** in the first state and the second state will be described in more detail, and the configuration in which two gaps G**1** and G**2** having different cross-sectional areas are formed on the nozzle **104** according to the operation of the needle valve **105** to control a spraying amount of an aerosol will be described in detail.

[0055] FIG. **4**A shows the inhaler **10** in the first state and FIG. **4**B shows the inhaler **10** in the second state.

[0056] In the first state, the nozzle **104** and the needle valve **105** may come into contact with each other and the mouthpiece **102** and the reservoir **103** may be isolated from each other.

[0057] In the second state, the needle valve **105** may move in the second direction so that the nozzle **104** and the needle valve **105** may not come into contact with each other, and the mouthpiece **102** and the reservoir **103** may communicate with each other through the nozzle **104**. Accordingly, the inhalable composition in the reservoir **103** may be discharged outside from the reservoir **103** through the mouthpiece **102**.

[0058] Specifically, the nozzle **104** may include a first nozzle portion **1041** and a second nozzle portion **1042**.

[0059] The first nozzle portion **1041** may be a portion adjacent to the mouthpiece **102**.

[0060] The second nozzle portion **1042** may be, for example, a portion located on the bottom of the first nozzle portion **1041** and adjacent to the reservoir **103**.

[0061] The nozzle **104** may be formed such that the first nozzle portion **1041** has a diameter smaller than that of the second nozzle portion **1042**.

[0062] The needle valve **105** may include a first valve portion **1051** and a second valve portion **1052**.

[0063] The first valve portion **1051** may be a portion formed at a front end of the needle valve **105**. The first valve portion **1051** may move adjacent to the first nozzle portion **1041** or the second nozzle portion **1042** when the piston **106** moves vertically. For example, in the first state, the first valve portion **1051** may be disposed adjacent to the first nozzle portion **1041**. In addition, in the second state, the first valve portion **1051** may move in the second direction to be adjacent to the second nozzle portion **1042**.

[0064] The second valve portion **1052** may be, for example, a portion formed below the first valve portion **1051**. A lower end of the second valve portion **1052** may be coupled to one surface of the piston **106**. Accordingly, when the piston **106** vertically reciprocates, the needle valve **105** may vertically reciprocate.

[0065] The needle valve **105** may be formed such that the first valve portion **1051** has a diameter smaller than that of the second valve portion **1052**.

[0066] In addition, the first valve portion **1051** may be formed to have a diameter smaller than that of the first nozzle portion **1041**. That is, the first valve portion **1051** may not come into contact with any of the first nozzle portion **1041** and the second nozzle portion **1042** in the first state or the second state.

[0067] Meanwhile, the nozzle **104** may further include a sealing member **1043** provided on the second nozzle portion **1042**. The sealing member **1043** may be provided as, for example, an O-ring or a Quad-ring formed of an elastic material such as silicon or rubber. An inner diameter of the sealing member **1043** may be larger than the diameter of the first valve portion **1051** and may be equal to or smaller than the diameter of the second valve portion **1052**. Also, the inner diameter of the sealing member **1043** may be larger than the diameter of the first nozzle portion **1041**.

[0068] The second valve portion **1052** may be disposed adjacent to the sealing member **1043** in the first state. That is, in the first state, the second valve portion **1052** may come into contact with the sealing member **1043**. Accordingly, the reservoir **103** may be sealed airtightly so that no aerosol leaks through the nozzle **104**.

[0069] On the other hand, the second valve portion **1052** may move in the second direction with respect to the sealing member **1043** and may not to come into contact with the nozzle **104** in the second state. Accordingly, the inhalable composition stored in the reservoir **103** may be discharged to the mouthpiece **102** through the nozzle **104**.

[0070] Referring to FIG. **4**B, a gap formed between the first nozzle portion **1041** and the first valve portion **1051** is defined as a first gap G**1**, and a gap formed between the sealing member **1043** and the first valve portion **1051** is defined as a second gap G**2**. Due to the structure of the nozzle **104** and the needle valve **105** described above, the first gap G**1** may be formed to be smaller than the second gap G**2**.

[0071] The reason for forming the diameter of the needle valve **105** differently according to the section and spatially dividing the nozzle **104** so that the first gap G**1** and the second gap G**2** having different cross-sectional areas are formed as described above is to uniformly control the spraying amount of the aerosol.

[0072] The spraying amount of the aerosol is generally proportional to the cross-sectional area of the nozzle **104**, and if the nozzle **104** is not spatially divided into the first gap G**1** and the second gap G**2**, the cross-sectional area of the nozzle **104** may be determined only by a gap between the sealing member **1043** and the needle valve **105**. At this time, when the sealing member **1043** provided as, for example, a Quad-ring is used, a tolerance of the Quad-ring is generally 0.05 mm, and thus, it is difficult to ensure a cross-sectional area for uniform spraying. Accordingly, the nozzle **104** may be spatially divided so that only the opening and closing of the inhalable composition is controlled in the second gap G**2**, and the aerosol is finally sprayed in the first gap G**1** where the tolerance is more easily controlled.

[0073] When the opening and spraying occur simultaneously, the spraying amount of the aerosol may be controlled only by a spraying area of the first gap G**1**.

[0074] Factors affecting the spraying amount may include pipe resistance, propulsion pressure, fluid viscosity, and the like, as well as the cross-sectional area of the nozzle described above.

[0075] Specifically, the spraying amount of the aerosol may vary depending on a distance that the inhalable composition moves to the first gap G**1**. When the second gap G**2** is the same, the spraying amount of the aerosol may vary depending on a stroke distance of the needle valve **105**. For example, when the stroke distance of the needle valve **105** is shortened, the distance through which the inhalable composition passes through the first gap G**1** may increase and the spraying amount may decrease.

[0076] The first gap G**1** may be formed in the first state or the second state. The second gap G**2** is formed when the first valve portion **1051** is disposed adjacent to the sealing member **1043**, and therefore, the second gap G**2** may be formed in the second state in which the needle valve **105** is moved in the second direction.

[0077] The second gap G**2** may allow the movement of the inhalable composition discharged from the reservoir **103**. The first gap G**1** may control a particle size of an aerosol which has passed through the second gap G**2**. That is, since the first gap G**1** is smaller than the second gap G**2**, the first gap G**1** may allow only the movement of an aerosol with fine particles from the aerosol. As a result, only an aerosol having a size that is able to pass through the first gap G**1** may move to the mouthpiece **102** to be discharged to the user.

[0078] That is, when the nozzle **104** is opened by a suction force, the aerosol is finally sprayed to the outside through the first gap G**1**, but the initial discharge from the reservoir **103** may be performed through the second gap G**2** formed between the sealing member **1043** and the first valve portion **1051**.

[0079] That is, since both the nozzle **104** and the needle valve **105** are formed straight, the first gap G**1** is constantly formed regardless of the opening and closing of the nozzle **104**, and therefore, the opening and closing is substantially performed between the sealing member **1043** and the needle valve **105**.

[0080] In this case, the first gap G**1** may be formed to have a size of 0.015 millimeters (mm) to 0.03 mm for the spraying of an aerosol with sufficiently small particles. For example, when the first gap G**1** is smaller than 0.015 mm, it is difficult for a liquid composition to move along a narrow gap, and thus, a gas may be mainly sprayed, and some liquid droplets may be weakly sprayed as if it is boiling. On the other hand, when the first gap G**1** is larger than 0.03 mm, large liquid droplets are mainly sprayed, which may cause an extremely large spraying amount per unit time, and the user may feel that the nasal cavity or oral cavity is wet. In addition, when the second gap G**2** is formed too narrow due to a manufacturing tolerance of the sealing member **1043**, a phenomenon which is the same phenomenon occurring due to the narrow first gap G**1** may occur, and therefore, in consideration of this point, the second gap G**2** may be formed sufficiently larger than the first gap G**1**. Accordingly, the inhaler **10** according to an embodiment may spray an aerosol with fine particles and easily adjust the spraying amount.

[0081] As described above, in the inhaler **10** according to an embodiment, the first gap G**1** and the second gap G**2** having different cross-sectional areas may be divided on the nozzle **104** for ejecting the aerosol according to the opening and closing operation of the needle valve **105**, and an appropriate section of the two divided gaps G**1** and G**2** is specified to enable uniform ejection of a sprayed aerosol.

[0082] In addition, as the aerosol is sprayed in a narrower section compared to the injection area by the first gap G**1** and the second gap G**2** having the donut shape, finer particles may be formed and a phenomenon of wetting or hitting the inside of the user's mouth may be improved.

[0083] In addition, the inhaler **10** according to an embodiment may maintain a state in which the nozzle **104** is closed by receiving a force of the needle valve **105** normally pushing up in the first direction due to the preload of the spring **107**. However, in the inhaler **10** according to an embodiment, when a suction force is applied through the mouthpiece **102** by the user, a negative pressure may be formed on the bottom of the piston **106**, the piston **106** may move downward by overcoming the preload of the spring **107**, and the needle valve **105** connected to the piston **106** may move downward, thereby forming the second gap G**2** between the sealing member **1043** and the needle valve **105**. Accordingly, the inhalable composition in the reservoir **103** may be discharged to the outside, for example, to the user's oral cavity through the first gap G**1** by passing through the second gap G**2** in the form of an aerosol. When the user stops inhaling, the piston **106** and the needle valve **105** may move upward again due to a restoring force of the spring **107** and the nozzle **104** may be closed to stop the spraying of the aerosol.

[0084] While the embodiments of the present disclosure have been described above with reference to specific components, and limited embodiments and drawings, the above descriptions are merely for better understanding of the present disclosure, and it will be apparent to one of ordinary skill in the art that various changes in form and details may be made in these embodiments without departing from the spirit and scope of the claims and their equivalents. For example, suitable results may be achieved if the described techniques are performed in a different order and/or if components in a described system, architecture, device, or circuit are combined in a different manner and/or replaced or supplemented by other components or their equivalents. Accordingly, the scope of the present disclosure is defined not by the detailed description, but by the claims and their equivalents, and all variations within the scope of the claims and their equivalents are to be construed as being included in the disclosure.

1. An inhaler comprising:
a housing having one surface, the other surface opposite to the one surface, and a plurality of side surfaces connecting the one surface and the other surface;
a mouthpiece which is disposed on the one surface of the housing;
a reservoir which is disposed inside the housing and stores an inhalable composition;
a nozzle which extends from the mouthpiece to the reservoir;
a needle valve which is movably disposed inside the nozzle and operates in a first state for closing the nozzle and a second state for opening the nozzle; and
a sealing member which is fixedly provided on the nozzle,
wherein, in the first state, a first gap is formed between the needle valve and the nozzle, and
wherein, in the second state, a second gap is formed between the needle valve and the sealing member.

2. The inhaler of claim **1**,
wherein the first state is defined as a state in which the needle valve comes into contact with the sealing member, and
wherein the second state is defined as a state in which the needle valve does not come into contact with the sealing member.

3. The inhaler of claim **1**, wherein the first gap is smaller than the second gap.

4. The inhaler of claim **1**,
wherein the nozzle comprises:
a first nozzle portion adjacent to the mouthpiece; and
a second nozzle portion adjacent to the reservoir, and
wherein the first nozzle portion is formed to have a diameter smaller than a diameter of the second nozzle portion.

5. The inhaler of claim **4**,
wherein the needle valve comprises:
a first valve portion which is formed on a front end and vertically reciprocates in a first direction from the other surface to the one surface of the housing or a second direction from the one surface to the other surface of the housing through the first nozzle portion and the second nozzle portion; and
a second valve portion which is formed on a lower portion of the first valve portion, and
wherein the first valve portion is formed to have a diameter smaller than a diameter of the second valve portion.

6. The inhaler of claim **5**, wherein the first valve portion is formed to have the diameter smaller than the diameter of the first nozzle portion.

7. The inhaler of claim **5**,
wherein the first valve portion does not come into contact with the first nozzle portion in the first state or the second state, and

wherein the second valve portion comes into contact with the sealing member in the first state and does not come into contact with the sealing member in the second state.

8. The inhaler of claim **5**,

wherein the first gap is formed between the first valve portion and the first nozzle portion, and

wherein the second gap is formed between the first valve portion and the sealing member.

9. The inhaler of claim **1**,

wherein the second gap allows movement of the inhalable composition discharged from the reservoir, and

wherein the first gap allows the movement of the inhalable composition to the mouthpiece by controlling a particle size of the inhalable composition passing through the second gap.

10. The inhaler of claim **1**, wherein the first gap is formed to 0.015 millimeters (mm) to 0.03 mm.

11. The inhaler of claim **1**, further comprising:

a piston which has one surface coupled to the needle valve, and vertically reciprocates in a first direction from the other surface to the one surface of the housing or a second direction from the one surface to the other surface of the housing; and

a spring which is provided below the piston in a preloaded state,

wherein, when a suction force is not applied through the mouthpiece, the spring pushes the piston up in the first direction to maintain the first state, and

wherein, when a suction force is applied through the mouthpiece, the piston overcomes preload of the spring and moves in the second direction to switch the state into the second state.

12. The inhaler of claim **11**, further comprising:

a passage which is formed inside the housing and extends from one side of the mouthpiece to a bottom portion of the piston,

wherein the passage transfers the suction force applied to the mouthpiece to the piston.

13. The inhaler of claim **12**, further comprising:

a negative pressure forming portion which forms a space between the piston and the passage,

wherein the negative pressure forming portion induces the piston to move in the second direction by generating a negative pressure by the suction force.

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